Rohit Mina

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.28
MATRIX HIGH SCHOOL , SIKAR	Class XII (Rajasthan Board of Secondary Education)	2022	76.40%
GSSS BAMBU , CHURU	Class X (Rajasthan Board of Secondary Education)	2020	77.17%

PROJECTS

CodeCollab -- Real-Time Collaborative Code Editor

Next.js · TypeScript · Node.js · PostgreSQL · Yjs · Socket.IO · Docker · Gemini · Azure

Live: https://app.codecollabapp.me

- Built a full-stack collaborative IDE supporting real-time multi-user editing, live presence, in-room chat, authentication, role-based membership, and persistent project storage.
- Engineered CRDT-based concurrent editing with Yjs and Monaco Editor, ensuring conflict-free, low-latency synchronization of editor state across simultaneous users.
- Designed a secure Docker-based code execution engine for Python, JavaScript, C++, and Java, enforcing container isolation, CPU/memory/PID limits, execution timeouts, output caps, and network lockdown.
- Integrated a project-aware Gemini AI assistant with selected-code and file/project context, powering Explain/Fix workflows and an in-editor AI chat.
- Implemented version history with snapshot creation, diff-style historical previews, and one-click restore, alongside file management and permission controls.
- Deployed a production-grade multi-service stack (frontend, backend, PostgreSQL, Yjs) on Azure using Docker Compose and Caddy with HTTPS on custom domains

SyntaxSeeker -- Agentic Knowledge Retrieval System

TypeScript · Node.js · LangChain · Gemini API · MemoryVectorStore · GitHub Actions

- Built an agentic RAG CLI enabling semantic search over codebases and technical docs using LangChain and Gemini.
- Engineered a hallucination-resistant pipeline -- document chunking, vector embeddings (text-embedding-004), and zero-temperature Gemini prompting for enterprise-grade reliability.
- Automated build validation via GitHub Actions CI, ensuring consistent code quality across the pipeline.

SKILLS

Languages: SQL, Python, JavaScript, TypeScript, C, Bash scripting

Cloud & DevOps: Microsoft Azure, Docker, Docker Compose, Caddy, CI/CD basics, Git/GitHub

Web & Backend: Node.js, Next.js, PostgreSQL, Socket.IO, REST APIs, WebSockets

AI/GenAI: Gemini API integration, prompt/context engineering, LLM-powered application design, AI-assisted development

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Ensemble Learning, Model Evaluation

Libraries & Tools: NumPy, Pandas, Matplotlib, Jupyter Notebook, Google Colab, VS Code

Concepts: OOP, Data Structures & Algorithms, System Design basics, Version Control workflows

POSITIONS OF RESPONSIBILITY

Coordinator, ELAN & Vision -- *2024-25*

- Led and mentored an *18-member deputy-coordinator team* to plan and *execute 15+ events* for ELAN & Vision, India's *largest techno-cultural fest*, hosting *~30,000 attendees.*
- Owned end-to-end planning and execution for *"Under 25,"* a flagship event *drawing 2,000-5,000* attendees -- managing logistics, scheduling, vendor coordination, and real-time troubleshooting.

Deputy Coordinator, ELAN & Vision -- *2023-24*

- Coordinated logistics and on-ground execution as part of the core organizing team for the same fest.

Event Coordinator, Farewell 2025

- Planned and managed the farewell event end-to-end for the graduating batch, coordinating logistics and vendors.

RELEVANT COURSES

Computer / AI & ML - Data Structure, Computational Physics, Introduction to AI, Introduction to Programming.

Mathematics - Advanced Mathematical Physics, Introduction to Probability, Introduction to Statistics, Differential Equation, Calculus - I, Calculus - II, Linear Algebra

Physics - Optics and Photonics, Special Relativity, Electronic Device Physics, Classical Mechanics

Other - Analog Electronics, Digital Electronics, Introduction to Entrepreneurship, Basic Electrical Engineering

EXTRACURRICULAR

- **Winner**, Inter-Hostel Volleyball Championship (Milan) -- Sept 2025, demonstrating teamwork under pressure
- **Winner**, Diesta Kabaddi Tournament -- inter-team championship showcasing agility and collaboration
- **Competitive athlete**: cricket, football, and 100m/200m sprints across inter-hostel and inter-departmental meets
- Actively participate in college-wide sports fests, including *Milan and Diesta*

HOBBIES AND INTERESTS

- *Avid follower of football and cricket; enjoy playing physical and competitive sports in free time*
- *Avid reader of self-development and psychology books, driven by curiosity for personal growth and human behavior*